

Mureşan Raul-Călin

raulmuresancalin@gmail.com | +40 752 140 363 |

[Personal Website](#) | [LinkedIn](#) | [GitHub](#)

EXPERIENCE

Data Analyst (Working Student) | Robert Bosch

Oct 2025 – Present

- **Engineered** an automated reminder system replacing manual follow-ups, reducing administrative workload for training compliance by **96%**.
- **Developed** a scalable data model using advanced DAX measures to map hierarchical structures across 3 management tiers.
- **Built** a real-time dashboard for training tracking, saving **~180 hours annually** by automating manual reporting tasks.

Programming Tutor | IT Junior

Jun 2024 – Sep 2025 (1y 4m)

- **Mentored** 22 students in C, C++, HTML, CSS and Python, adapting complex curriculum to individual learning speeds and skill levels.
- **Designed** practical projects (Basic AI implementations, games, websites) to increase student engagement and technical retention.

EDUCATION

Technical University of Cluj-Napoca | B.Sc. in Computer Science

2022 – 2026 (Graduated)

- **Related Coursework:** Data Structures & Algorithms, Machine Learning, OOP, Artificial Intelligence.

PROJECTS

[MathSim](#) - Automated Grid Test Digitization & Grading (Bachelor's Thesis) | Python, OpenCV, Tesseract OCR, Next.js

- **Built** a full-stack platform that automatically ingests, digitizes, and grades a 166-page math test dataset (959 questions).
- **Engineered** a CV pipeline: contour-based segmentation, Page Sequence Voting for OCR indexing, and answer extraction via morphological dilation.
- **Optimized** image processing pipeline using multiprocessing, achieving 7.2× speedup (reduced execution time for the full dataset from 63s to 8.7s).
- **Developed** a FastAPI backend (JWT, weighted simulation generation, Ollama LLM integration for user progress feedback) and a Next.js 16 frontend (quiz player with timer, stats).

[NoteFlow](#) | React, Next.js, Convex, TypeScript, Tailwind CSS

- **Developed** a content management app optimizing load times and SEO via Next.js Server-Side Rendering (SSR).
- **Built** a type-safe, modular architecture using TypeScript to ensure scalability and reduce runtime errors.

[Questify](#) | React, PostgreSQL, REST API

- **Designed** a normalized PostgreSQL database schema to efficiently handle complex user interactions and voting logic.
- **Implemented** secure RESTful endpoints ensuring data integrity for a Q&A platform aimed at technical knowledge exchange.

[IoT Security System](#) | C++, Arduino UNO R4, IFTTT

- **Created** a motion-detection system integrating temperature/humidity sensors and real-time alerts.
- **Configured** IFTTT webhooks to trigger immediate email notifications based on specific sensor data thresholds.

[Pacman](#) | Assembly x86

- **Engineered** a low-level game clone focusing on manual memory management and optimized graphics rendering logic.

SKILLS

Programming Languages: Python, C/C++, JavaScript/TypeScript, SQL, Assembly x86

Technologies: React, Next.js, PostgreSQL, Tailwind CSS, OpenCV, Arduino, Git, Linux

Languages: Romanian (Native), English (Fluent), French (Intermediate)

PROGRAMMING CONTESTS

Cloudflight Coding Contest (CCC) – 2023, 2024, 2025